# Chapter 11. Groups

*Subgroups, homomorphisms, cyclic and linear groups*

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## 11.1 Binary operations and groups

**Motivation.** Vector addition, matrix multiplication and composition of rigid motions share associativity, identity and inverses: a group.

$X \times Y = \{(x, y) : x \in X, y \in Y\}$ is the **Cartesian product**; $(x, y)$ is an **ordered pair**.

**Definition 11.1.1 (Binary operation)**

A function $* : X \times X \to X$ is a **binary operation** on $X$; we write $x * y$, or $x \cdot y$, $xy$.

**Definition 11.1.2 (Associativity)**

A binary operation is **associative** if $(xy)z = x(yz)$ for all $x, y, z \in X$; then $xyz$ is meaningful.

**Example 11.1.3**

On $\mathfrak{M}_{n,n}(F)$, the bracket $[A, B] = AB - BA$ is not associative: for $n = 2$, $[[E_{12}, E_{21}], E_{21}] = [E_{11} - E_{22}, E_{21}] = -2E_{21}$, while $[E_{12}, [E_{21}, E_{21}]] = 0$.

From now on every binary operation is assumed associative.

**Definition 11.1.4 (Group)**

A set $G$ with a binary operation is a **group** if (G1) $(xy)z = x(yz)$ for all $x, y, z \in G$; (G2) there is $e \in G$ with $xe = x = ex$ for all $x \in G$; (G3) for each $x \in G$ there is $\tilde{x} \in G$ with $x\tilde{x} = e = \tilde{x}x$.

**Example 11.1.5**

(a) $(\{-1, 1\}, +)$ is not a group ($1 + 1 \notin \{-1, 1\}$), but $(\{-1, 1\}, \times)$ is.
(b) $\{e\}$ with $ee = e$ is the **trivial group**.
(c) $(\mathbb{Z}, +)$ is a group; $(\mathbb{Z}, \times)$ is not.
(d) $(F, +)$ and $(F - \{0\}, \times)$ are groups; $(F, \times)$ is not.
(e) For a vector space $V$, $(V, +)$ is a group (axioms (V1)–(V4)).
(f) The symmetric group $(S_n, \circ)$ of Chapter 6 is a group.

**Observation 11.1.6**

A group is not empty.

**Proof.** By (G2) it contains $e$. ■

**Observation 11.1.7**

The element $e$ of (G2) is unique; it is the **identity** of $G$, written $e_G$.

**Proof.** If $e'$ is another, $e = ee' = e'$. ■

**Observation 11.1.8**

For each $x \in G$, the $\tilde{x}$ of (G3) is unique; it is the **inverse** $x^{-1}$.

**Proof.** If $xx' = e = x'x$ as well, then $\tilde{x} = \tilde{x}e = \tilde{x}(xx') = (\tilde{x}x)x' = ex' = x'$. ■

**Exercise 11.1.9**

For $g, h \in G$: (a) $(gh)^{-1} = h^{-1}g^{-1}$;
(b) $(g^{-1})^{-1} = g$.

**Solution.** (a) $(gh)(h^{-1}g^{-1}) = g(hh^{-1})g^{-1} = e$ and likewise on the other side; use uniqueness of inverses.

(b) $g$ satisfies the defining property of the inverse of $g^{-1}$. □

**Definition 11.1.10 (Order)**

$G$ is a **finite group** if it is a finite set; then $|G|$ is its **order**. Otherwise $G$ is an **infinite group**.

**Definition 11.1.11 (Commutative group)**

$G$ is **commutative** (**abelian**) if $xy = yx$ for all $x, y \in G$.

**Exercise 11.1.12**

$\mu_n = \{\alpha \in \mathbb{C} : \alpha^n = 1\}$, under multiplication of complex numbers, is a finite commutative group of order $n$.

**Solution.** $\mu_n$ contains 1 and is closed under products and inverses; associativity and commutativity come from $\mathbb{C}$. Its elements are the $n$ distinct $e^{2\pi ik/n}$, $0 \le k < n$ (at most $n$ roots of $t^n - 1$). □

**Exercise 11.1.13**

For $n \ge 3$, $S_n$ is a finite non-commutative group of order $n!$.

**Solution.** $|S_n| = n!$. $(1\,2)(1\,3)$ sends $1 \mapsto 3 \mapsto 3$ but $(1\,3)(1\,2)$ sends $1 \mapsto 2$; so they differ. □

**Exercise 11.1.14**

On $F^3$ define $(a_1, a_2, a_3)^t \bullet (b_1, b_2, b_3)^t = (a_1 + b_1,\ a_2 + b_2 + a_1b_3,\ a_3 + b_3)^t$. Then $(F^3, \bullet)$ is an infinite non-commutative group.

**Solution.** Both $(a\bullet b)\bullet c$ and $a\bullet(b\bullet c)$ equal $(a_1+b_1+c_1,\ a_2+b_2+c_2+a_1b_3+a_1c_3+b_1c_3,\ a_3+b_3+c_3)$. The identity is 0, and the inverse of $a$ is $(-a_1, -a_2+a_1a_3, -a_3)$. Non-commutative: $e_1 \bullet e_3 = (1, 1, 1)$ but $e_3 \bullet e_1 = (1, 0, 1)$. (This is the Heisenberg group of upper unitriangular $3\times3$ matrices in disguise.) □

## 11.2 Elementary properties of groups

**Motivation.** Powers, cancellation, and multiplication tables classify the smallest groups.

**Notation 11.2.1**

(a) (Multiplicative notation) The identity is 1; $g^1 = g$, $g^2 = gg$, ..., $g^0 = 1$, $g^{-2} = g^{-1}g^{-1}$, ....
(b) (Additive notation) For an abelian group $A$ written with $+$: identity 0, inverse $-a$, and $\ldots, (-2)a = (-a) + (-a)$, $0a = 0$, $1a = a$, $2a = a + a, \ldots$; $a - b = a + (-b)$.
(c) Commutative groups are usually written multiplicatively and abelian groups additively.
(d) Unless stated otherwise we use multiplicative notation.

**Exercise 11.2.2**

(a) $(g^n)^{-1} = (g^{-1})^n = g^{-n}$ for $n \in \mathbb{Z}$.
(b) Additively: $-(na) = n(-a) = (-n)a$.

**Solution.** $g^n(g^{-1})^n = 1$ by cancelling adjacent $gg^{-1}$; and $g^{-n} = (g^{-1})^n$ by definition (for $n > 0$; the other cases follow). □

**Exercise 11.2.3**

$(h^{-1}gh)^n = h^{-1}g^nh$ for $n \in \mathbb{Z}$.

**Solution.** For $n > 0$, the inner $hh^{-1}$ cancel; $n = 0$ is clear; for $n < 0$ apply this to $g^{-1}$, using $(h^{-1}gh)^{-1} = h^{-1}g^{-1}h$. □

**Exercise 11.2.4**

Elements $ghg^{-1}h^{-1}$ are **commutators**. $G$ is commutative iff all commutators are 1.

**Solution.** $ghg^{-1}h^{-1} = 1 \iff gh = hg$. □

**Exercise 11.2.5**

If $g^2 = 1$ for all $g \in G$, then $G$ is commutative.

**Solution.** Then $g = g^{-1}$ for all $g$, so $ghg^{-1}h^{-1} = ghgh = (gh)^2 = 1$. □

**Exercise 11.2.6 (Exponent laws, multiplicative)**

For $g, h \in G$, $m, n \in \mathbb{Z}$: (a) $g^m g^n = g^{m+n}$;
(b) $(g^m)^n = g^{mn}$;
(c) if $G$ is commutative, $g^n h^n = (gh)^n$.

**Solution.** For $m, n \geq 0$ count factors; negative exponents reduce to this via $g^{-k} = (g^{-1})^k$. (c): regroup $(gh)\cdots(gh)$ by commutativity. □

**Exercise 11.2.7 (Exponent laws, additive)**

For $a, b \in A$, $m, n \in \mathbb{Z}$: (a) $ma + na = (m + n)a$;
(b) $m(na) = (mn)a$;
(c) $na + nb = n(a + b)$.

**Solution.** Exercise 11.2.6 in additive notation. □

**Observation 11.2.8 (Cancellation law)**

If $xy = xz$, then $y = z$; if $xz = yz$, then $x = y$.

**Proof.** $y = (x^{-1}x)y = x^{-1}(xy) = x^{-1}(xz) = z$; the other side is symmetric. ■

**Exercise 11.2.9**

State Observation 11.2.8 additively.

**Solution.** If $a + b = a + c$ then $b = c$; if $a + c = b + c$ then $a = b$. □

**Example 11.2.10**

$xy = x$ implies $y = 1$, since $xy = x1$.

**Exercise 11.2.11**

(a) If $y = u^{-1}xu$, then $x = uyu^{-1}$.
(b) If $u^{-1}xu = 1$, then $x = 1$.

**Solution.** (a) Multiply by $u$ on the left and $u^{-1}$ on the right.

(b) By (a), $x = u1u^{-1} = 1$. □

A finite group $G = \{1, x_2, x_3, \dots\}$ can be described by its **multiplication table**, whose $(x_i, x_j)$ entry is $x_i x_j$.

**Example 11.2.12**

All groups of order 1 are "the same": $\{e\}$ and $(\{0\}, +)$ differ only in names.

**Example 11.2.13**

If $G = \{1, x\}$, then $x^2 \neq x$ (cancellation), so $x^2 = 1$: the table is forced, and $G$ is $\mu_2$ up to renaming.

**Observation 11.2.14**

In a multiplication table every row contains each element exactly once, and so does every column.

**Proof.** $xy = xz$ forces $y = z$ (cancellation), so a row has no repetitions; $G$ is finite, so each element appears. Similarly for columns. ■

**Observation 11.2.15**

For fixed $x \in G$, $\lambda_x : G \to G$, $\lambda_x(y) = xy$, is a bijection (it permutes $G$).

**Proof.** Injective by cancellation; surjective since $z = \lambda_x(x^{-1}z)$. ■

**Example 11.2.16**

If $G = \{1, x, y\}$, then $x^2 \neq x$, and $x^2 = 1$ would force $xy = y$, contradicting cancellation; so $x^2 = y$, and by Observation 11.2.14 the table is

| | $1$ | $x$ | $y$ |
|---|---|---|---|
| $1$ | $1$ | $x$ | $y$ |
| $x$ | $x$ | $y$ | $1$ |
| $y$ | $y$ | $1$ | $x$ |

So there is at most one group of order 3 up to renaming.

**Question 11.2.17**

How can we check that a given table is associative?

If a known group realises the table, associativity is automatic: for $\zeta = e^{2\pi i/3}$, the renaming $x \mapsto \zeta$, $y \mapsto \zeta^2$ carries the table of Example 11.2.16 to that of $\mu_3$.

**Example 11.2.18**

For $G = \{1, x, y, z\}$, cancellation and Observation 11.2.14 leave four possible tables (up to symmetry): one in which every element squares to 1, and three in which one non-identity element squares to 1 (the second, third and fourth tables), which are $\mu_4 = \{1, -1, i, -i\}$ up to renaming.

**Exercise 11.2.19**

The second, third and fourth tables of Example 11.2.18 are associative.

**Solution.** Each is the table of $\mu_4$ after a renaming (e.g. $x \mapsto -1$, $y \mapsto i$, $z \mapsto -i$). □

**Example 11.2.20**

For groups $G_1, G_2$, the Cartesian product $G_1 \times G_2$ with $(g_1, g_2)(h_1, h_2) = (g_1h_1, g_2h_2)$ is a group, the **direct product**.

**Example 11.2.21**

$\mu_2 \times \mu_2$ realises the first table of Example 11.2.18; e.g. $(1, -1)(1, -1) = (1, 1)$. It is the **Klein 4-group**, not $\mu_4$ (all its elements square to 1).

**Exercise 11.2.22**

(If you can,) show with tables that a group of order 5 is $\mu_5$ up to renaming.

**Solution [Supplement] (via Lagrange's theorem, Observation 12.1.10).** Let $x \neq 1$. The powers of $x$ form a subgroup $\langle x \rangle$ (§11.3) whose order divides 5, hence equals 5. So $G = \{1, x, x^2, x^3, x^4\}$ with $x^5 = 1$, and $x^k \mapsto e^{2\pi ik/5}$ is a renaming onto $\mu_5$ preserving the table. □

**Exercise 11.2.23**

If $|G| \leq 5$, then $G$ is commutative; so $S_3$ is the smallest non-commutative group.

**Solution.** By Examples 11.2.12, 11.2.13, 11.2.16, 11.2.18 and Exercise 11.2.22, every such group is $\mu_n$ or $\mu_2 \times \mu_2$, whose tables are symmetric. $S_3$ has order 6 and is non-commutative. □

**Remark 11.2.24**

Semigroups, monoids, and one-sided identities and inverses appear in other texts; they are not needed here, nor is the classification of finite groups.

## 11.3 Subgroups

**Motivation.** Subgroups play the role of subspaces; the subgroup generated by $S$ consists of words in $S \cup S^{-1}$.

**Definition 11.3.1 (Subgroup)**

A subset $H \subseteq G$ that is a group under the operation of $G$ is a **subgroup**, $H \leq G$.

**Observation 11.3.2**

Let $H \leq G$. (a) $e_H = e_G$.
(b) The inverse of $h \in H$ in $H$ equals its inverse $h^{-1}$ in $G$.

**Proof.** (a) In $G$, $e_H e_H = e_H = e_H e_G$; cancel $e_H$.

(b) In $G$, $hh' = e_H = e_G = hh^{-1}$; cancel $h$. ■

**Exercise 11.3.3**

Why was $0_W = 0_V$ for subspaces $W \leq V$ never an issue?

**Solution.** $0_W = 0 \cdot w = 0_V$ for any $w \in W$. □

**Observation 11.3.4**

$H \subseteq G$ is a subgroup iff (S1) $h_1, h_2 \in H \Rightarrow h_1 h_2 \in H$; (S2) $e \in H$; (S3) $h \in H \Rightarrow h^{-1} \in H$.

**Proof.** Necessity: the operation must restrict to $H$ (S1), and Observation 11.3.2 gives (S2), (S3). Sufficiency: associativity is inherited from $G$. ■

**Exercise 11.3.5**

A nonempty $H \subseteq G$ is a subgroup iff $h_1, h_2 \in H \Rightarrow h_1 h_2^{-1} \in H$.

**Solution.** Necessity is clear. For sufficiency, take $h \in H$: then $e = hh^{-1} \in H$, $h^{-1} = eh^{-1} \in H$, and $h_1 h_2 = h_1 (h_2^{-1})^{-1} \in H$. □

**Example 11.3.6**

(a) $\{e\} \leq G \leq G$. (b) Under addition, $\mathbb{Z} < \mathbb{Q} < \mathbb{R} < \mathbb{C}$. (c) Under multiplication, $\mu_2 < \mathbb{Q}-\{0\} < \mathbb{R}-\{0\} < \mathbb{C}-\{0\}$. (d) The alternating group $A_n$ is a subgroup of $S_n$. (e) If $m \mid n$, then $\mu_m \leq \mu_n$. (f) If $W$ is a subspace of $V$, then $(W,+) \leq (V,+)$.

**Observation 11.3.7**

If $K \leq H$ and $H \leq G$, then $K \leq G$.

**Proof.** $K$ is a group under the operation of $H$, which is that of $G$. ■

**Exercise 11.3.8**

If $I \neq \emptyset$ and $H_i \leq G$ for all $i \in I$, then $\bigcap_i H_i \leq G$.

**Solution.** Each $H_i$ contains $e$ and is closed under products and inverses; so is the intersection. □

**Notation 11.3.9**

For $H, K \subseteq G$, $HK = \{hk : h \in H, k \in K\}$.

**Exercise 11.3.10**

(a) If $G$ is commutative and $H, K \leq G$, then $HK \leq G$.
(b) In $S_3 = \{id, (1\,2), (1\,3), (2\,3), (1\,2\,3), (1\,3\,2)\}$, $H = \{id, (1\,2)\}$ and $K = \{id, (2\,3)\}$ are subgroups but $HK \not\leq S_3$.

**Solution.** (a) $(h_1k_1)(h_2k_2)^{-1} = h_1h_2^{-1}k_1k_2^{-1} \in HK$ by commutativity; apply Exercise 11.3.5. (b) $HK = \{id, (2\,3), (1\,2), (1\,2)(2\,3)\}$ does not contain $((1\,2)(2\,3))^{-1} = (2\,3)(1\,2)$. □

**Definition 11.3.11 (Generated subgroup)**

For $S \subseteq G$, the smallest subgroup of $G$ containing $S$ is the **subgroup generated by** $S$, $\langle S \rangle$: $S \subseteq \langle S \rangle \leq G$, and $S \subseteq H \leq G$ implies $\langle S \rangle \subseteq H$.

**Observation 11.3.12**

$\langle S \rangle = \bigcap_{S \subseteq H \leq G} H$; in particular $\langle S \rangle$ exists and is unique.

**Proof.** By Exercise 11.3.8 the intersection is a subgroup containing $S$, and it lies in every subgroup containing $S$. ■

**Example 11.3.13**

$\langle \emptyset \rangle = \{e\}$, and $\langle x \rangle = \{x^i : i \in \mathbb{Z}\}$ (a subgroup by the exponent laws); possibly $x^i = x^j$ for $i \neq j$.

**Definition 11.3.14 (Cyclic group)**

(a) $\langle x \rangle$ is the **cyclic subgroup** generated by $x$.
(b) If $G = \langle x \rangle$ for some $x$, $G$ is a **cyclic group** with generator $x$. Cyclic groups are commutative (§11.7).

**Definition 11.3.15 (Alphabet and words)**

(a) For $S \subseteq G$ let $S^{-1} = \{s^{-1} : s \in S\}$; $S \cup S^{-1}$ is the **alphabet**.
(b) Finite products of letters are **words**; $e$ is the empty word. (Different words may represent the same element.)

**Observation 11.3.16**

The set of all words in the alphabet $S \cup S^{-1}$ is a subgroup of $G$.

**Proof.** It contains the empty word, products of words are words, and $(s_1 \cdots s_k)^{-1} = s_k^{-1} \cdots s_1^{-1}$ is a word (Exercise 11.1.9); apply Observation 11.3.4. ■

**Proposition 11.3.17**

$\langle S \rangle$ is the set $K$ of all words in the alphabet $S \cup S^{-1}$.

**Proof.** $\langle S \rangle$ is a subgroup containing $S$, so it contains all such words: $K \subseteq \langle S \rangle$. Conversely $K$ is a subgroup (Observation 11.3.16) containing $S$, so $\langle S \rangle \subseteq K$. ■

**Exercise 11.3.18**

(a) $(\mathbb{Z}, +)$ is cyclic; how many generators does it have?
(b) $3\mathbb{Z} = \{3a : a \in \mathbb{Z}\}$ is a cyclic subgroup of $(\mathbb{Z}, +)$.
(c) $\langle 2, 3 \rangle = \mathbb{Z} = \langle 4, 15 \rangle$, but $\langle 6, 9 \rangle \neq \mathbb{Z}$.

**Solution.** (a) $\mathbb{Z} = \langle 1 \rangle = \langle -1 \rangle$, and these are the only generators: $\langle n \rangle = n\mathbb{Z} \neq \mathbb{Z}$ for $|n| \neq 1$.

(b) $3\mathbb{Z} = \langle 3 \rangle$.

(c) $1 = 3 - 2 = 4 \cdot 4 - 15$, so 1 lies in both; but every element of $\langle 6, 9 \rangle$ is a multiple of 3, so $\langle 6, 9 \rangle = 3\mathbb{Z}$. □

**Exercise 11.3.19**

(a) $\mu_5$ is cyclic; how many generators?
(b) $\mu_6$ is cyclic; how many generators?

**Solution.** $\mu_n = \langle \zeta \rangle$ with $\zeta = e^{2\pi i/n}$, and $\zeta^k$ generates iff $\gcd(k, n) = 1$. (a) 4 generators ($k = 1, 2, 3, 4$).

(b) 2 generators ($k = 1, 5$). □

**Exercise 11.3.20**

(a) Find all subgroups of $S_3$.
(b) Is $S_3$ cyclic?
(c) $\langle (1\,2), (2\,3) \rangle = S_3 = \langle (1\,2), (1\,2\,3) \rangle$.

**Solution.** (a) $\{id\}$, $\langle(1\,2)\rangle$, $\langle(1\,3)\rangle$, $\langle(2\,3)\rangle$, $A_3 = \langle(1\,2\,3)\rangle$, $S_3$ (two transpositions, or a transposition and a 3-cycle, generate $S_3$ by (c)).

(b) No: element orders are $1, 2, 3$.

(c) $(1\,2)(2\,3)$, $(2\,3)(1\,2)$ are the two 3-cycles and $(1\,2)(2\,3)(1\,2) = (1\,3)$. Likewise $(1\,2\,3)^2$, $(1\,2)(1\,2\,3)$, $(1\,2\,3)(1\,2)$ give the rest. □

## 11.4 The heart of undergraduate algebra

**Motivation.** Every subgroup of $(\mathbb{Z}, +)$ is $n\mathbb{Z}$, by division by its least positive element.

**Example 11.4.1**

For $n \in \mathbb{Z}$, $n\mathbb{Z} = \{na : a \in \mathbb{Z}\} = \langle n\rangle$ is a subgroup of $(\mathbb{Z}, +)$.

**Theorem 11.4.2 (The heart of undergraduate algebra)**

The subgroups of $(\mathbb{Z}, +)$ are exactly the $n\mathbb{Z}$, $0 \le n \in \mathbb{Z}$.

**Proof.** Let $0 \ne H \le \mathbb{Z}$ and $n$ its least positive element. For $a \in H$, $a = qn + r$ with $0 \le r < n$ gives $r = a - qn \in H$, so $r = 0$. Hence $H \le n\mathbb{Z} = \langle n\rangle \le H$. ($H = 0$ is $0\mathbb{Z}$.) ■

## 11.5 Group isomorphisms

**Motivation.** Isomorphic groups differ only in the names of their elements.

**Definition 11.5.1 (Isomorphism)**

Groups $G, G'$ are **isomorphic**, $G \approx G'$, if there is a bijection $\varphi : G \to G'$ with $\varphi(g_1 g_2) = \varphi(g_1)\varphi(g_2)$; $\varphi$ is a (**group**) **isomorphism**. Equivalently, with $(\varphi \times \varphi)(g_1, g_2) = (\varphi g_1, \varphi g_2)$, the square

$$\begin{array}{ccc} G \times G & \xrightarrow{*} & G \\ {\scriptstyle \varphi\times\varphi}\downarrow & & \downarrow{\scriptstyle \varphi} \\ G' \times G' & \xrightarrow[*']{} & G' \end{array}$$

commutes.

**Example 11.5.2**

Let $\varphi : G \to H$ be an isomorphism, $g \in G$, $S \subseteq G$. Then (a) $\varphi(e_G) = e_H$;
(b) $\varphi(g^{-1}) = \varphi(g)^{-1}$;
(c) if $\langle S\rangle = G$, then $\langle\varphi(S)\rangle = H$;
(d) if $g^2 = e_G$, then $\varphi(g)^2 = e_H$;
(e) if $G$ is commutative, so is $H$;
(f) if $G$ is cyclic, so is $H$;
(g) if $G$ has six subgroups, so does $H$.

**Proof [Supplement].** (a) $\varphi(e)\varphi(e) = \varphi(e) = \varphi(e)e_H$; cancel.

(b) $\varphi(g)\varphi(g^{-1}) = \varphi(e) = e_H$.

(c) $\varphi$ maps words in $S \cup S^{-1}$ onto words in $\varphi S \cup (\varphi S)^{-1}$ (Proposition 11.3.17) and is onto.

(d) Apply $\varphi$.

(e) $\varphi(a)\varphi(b) = \varphi(ab) = \varphi(ba) = \varphi(b)\varphi(a)$, and $\varphi$ is onto.

(f) Case of (c).

(g) $K \mapsto \varphi(K)$ is a bijection between the sets of subgroups (with inverse $K' \mapsto \varphi^{-1}(K')$). ■

**Exercise 11.5.3**

Is $\mathbb{Z} \approx \mathbb{Z} \times \mathbb{Z}$ (additive groups)?

**Solution.** No (Example 11.5.2(f)): $\mathbb{Z} \times \mathbb{Z}$ is not cyclic, as $\langle (a,b) \rangle = \{(na, nb)\}$ never contains both $(1,0)$ and $(0,1)$. □

**Example 11.5.4**

Properties *not* preserved by isomorphisms: "the operation is called multiplication", "$G$ has an element named $x$", "the elements are complex numbers", "$|g| = 7$ as a real number", "$g$ integrates to $h$".

**Observation 11.5.5**

$\approx$ is an equivalence relation.

**Proof.** Exercise 11.5.6. ■

**Exercise 11.5.6**

(a) $id_G$ is an isomorphism.
(b) If $\varphi : G \to H$ is an isomorphism, so is $\varphi^{-1}$.
(c) If $\psi : K \to G$, $\varphi : G \to H$ are isomorphisms, so is $\varphi \circ \psi$.

**Solution.** (b) With $h_i = \varphi(g_i)$: $\varphi^{-1}(h_1 h_2) = \varphi^{-1}(\varphi(g_1 g_2)) = g_1 g_2 = \varphi^{-1}(h_1)\varphi^{-1}(h_2)$. (a), (c) are direct. □

**Example 11.5.7**

$\exp : (\mathbb{R}, +) \to (\mathbb{R}^+, \times)$ is a group isomorphism, with inverse log: $e^{a+b} = e^a e^b$.

**Remark 11.5.8**

Unlike vector spaces (dimension), deciding whether two groups are isomorphic is in general hard.

## 11.6 Group homomorphisms

**Motivation.** Homomorphisms are the analogue of linear maps, with kernels and images.

**Definition 11.6.1 (Homomorphism)**

$\varphi : G \to H$ is a (**group**) **homomorphism** if $\varphi(xy) = \varphi(x)\varphi(y)$ for all $x, y \in G$.

**Observation 11.6.2**

For a homomorphism $\varphi$: (a) $\varphi(e) = e$;
(b) $\varphi(x^{-1}) = \varphi(x)^{-1}$.

**Proof.** As in Example 11.5.2(a), (b). ■

**Definition 11.6.3 (Kinds of homomorphisms)**

A homomorphism $\varphi : G \to H$ is a **monomorphism** if injective, an **epimorphism** if surjective, an **isomorphism** if bijective, an **endomorphism** if $G = H$, and an **automorphism** if it is a bijective endomorphism.

**Observation 11.6.4**

A homomorphism $\varphi : G \to H$ is an isomorphism iff there is a homomorphism $\psi : H \to G$ with $\psi\varphi = id_G$ and $\varphi\psi = id_H$.

**Proof.** If $\varphi$ is an isomorphism, take $\psi = \varphi^{-1}$ (Exercise 11.5.6). Conversely such a $\psi$ is an inverse function, so $\varphi$ is bijective. ■

**Definition 11.6.5 (Kernel and image)**

$\ker \varphi = \varphi^{-1}(e) = \{x \in G : \varphi(x) = e\}$ and $\operatorname{im} \varphi = \varphi(G)$.

**Observation 11.6.6**

For a homomorphism $\varphi : G \to H$: (a) $\ker \varphi \leq G$ and $\operatorname{im} \varphi \leq H$;
(b) if $\varphi$ is a monomorphism, $G \approx \operatorname{im} \varphi$;
(c) if $K \leq G$, then $\varphi(K) \leq H$.

**Proof [Supplement].** (a) If $\varphi x = \varphi y = e$, then $\varphi(xy^{-1}) = e$, and $e \in \ker \varphi$ (Exercise 11.3.5).

(c) $\varphi(x)\varphi(y)^{-1} = \varphi(xy^{-1}) \in \varphi(K)$; the case $K = G$ gives $\operatorname{im} \varphi \leq H$.

(b) $\varphi : G \to \operatorname{im} \varphi$ is a bijective homomorphism. ■

**Observation 11.6.7**

For a homomorphism $\varphi$ the following are equivalent: (1) $\varphi$ is a monomorphism; (2) $\varphi x = \varphi y \Rightarrow x = y$; (3) $\varphi x = \varphi e = e \Rightarrow x = e$; (4) $\ker \varphi = \{e\}$.

**Proof.** All but (3)⇒(2) are immediate. If (3) holds and $\varphi x = \varphi y$, then $\varphi(xy^{-1}) = \varphi(x)\varphi(y)^{-1} = e$, so $xy^{-1} = e$, $x = y$. ■

**Observation 11.6.8**

For a homomorphism $\varphi : G \to H$ and $S \subseteq G$: (a) $\varphi\langle S\rangle = \langle \varphi S\rangle$;
(b) if $G = \langle S\rangle$, then $\operatorname{im} \varphi = \langle \varphi S\rangle$;

(c) $\varphi$ is an epimorphism iff it maps a generating set of $G$ to a generating set of $H$.

**Proof.** (a) Images of words in $S \cup S^{-1}$ are words in $\varphi S \cup (\varphi S)^{-1}$ (Proposition 11.3.17), so $\varphi\langle S\rangle \leq \langle \varphi S\rangle$; conversely $\varphi\langle S\rangle \leq H$ contains $\varphi S$ (Observation 11.6.6(c)).

(b) The case $\langle S\rangle = G$.

(c) By (b), $\operatorname{im}\varphi = H$ iff $\langle \varphi S\rangle = H$. ■

**Exercise 11.6.9**

If $\varphi : G \to H$ is a homomorphism and $G$ is cyclic, then $\operatorname{im}\varphi$ is cyclic.

**Solution.** If $G = \langle x\rangle$, then $\operatorname{im}\varphi = \langle \varphi x\rangle$ by Observation 11.6.8(b). □

**Exercise 11.6.10**

Let $\varphi(g) = g^{-1}$ and $\psi(g) = g^2$ on $G$. (a) $\varphi$ is a bijection; is it a homomorphism?
(b) Is $\psi$ a homomorphism?

**Solution.** (a) $\varphi = \varphi^{-1}$; it is a homomorphism iff $y^{-1}x^{-1} = x^{-1}y^{-1}$ for all $x, y$, iff $G$ is commutative.

(b) $xyxy = xxyy$ iff $yx = xy$: again iff $G$ is commutative. □

**Exercise 11.6.11**

(a) For $n \geq 2$, $\operatorname{sgn} : S_n \to \mu_2$ is an epimorphism.
(b) Describe $\ker(\operatorname{sgn})$.

**Solution.** (a) $\operatorname{sgn}(\sigma\tau) = \operatorname{sgn}\sigma \operatorname{sgn}\tau$, and $\operatorname{sgn}(1\,2) = -1$.

(b) $\ker(\operatorname{sgn}) = A_n$, the even permutations. □

## 11.7 Cyclic groups

**Motivation.** Up to isomorphism a cyclic group is $\mathbb{Z}$ or $\mu_n$.

**Example 11.7.1**

(a) $\mathbb{Z} = \langle 1\rangle$ is an infinite cyclic group, and so is $n\mathbb{Z} = \langle n\rangle$ for $n > 0$; $a \mapsto na$ is an isomorphism $\mathbb{Z} \approx n\mathbb{Z}$ onto a proper subgroup.
(b) For $\zeta = e^{2\pi i/n}$, $\mu_n = \{1, \zeta, \ldots, \zeta^{n-1}\} = \langle \zeta\rangle$ is cyclic of order $n$.

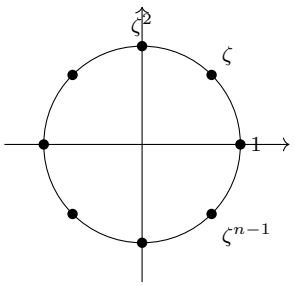



**Figure.** *$\mu_n$ ($n = 8$): the vertices of a regular $n$-gon on the unit circle.*

**Definition 11.7.2 (Order of an element)**

The **order** of $x \in G$ is $|x| = |\langle x \rangle|$.

**Observation 11.7.3**

If $x$ has finite order, then $|x| = n$ iff (1) $x^n = 1$ and (2) $x^m \neq 1$ for $0 < m < n$; i.e. $n$ is the least positive integer with $x^n = 1$.

**Proof.** $\langle x \rangle$ is finite, so $x^i = x^j$ for some $i < j$: $\{a \geq 1 : x^a = 1\} \neq \emptyset$; let $n$ be its least element (the $n$ of (1), (2)). For $b = qn + r$, $0 \leq r < n$, $x^b = (x^n)^q x^r = x^r$, so $\langle x \rangle = \{1, x, \ldots, x^{n-1}\}$; these are distinct, since $x^k = x^\ell$, $0 \leq k < \ell < n$, gives $x^{\ell-k} = 1$. So $|x| = n$. ■

**Definition 11.7.4 (Order, second definition)**

For $x \in G$, $\gamma_x : \mathbb{Z} \to G$, $\gamma_x(a) = x^a$, is a homomorphism (exponent laws), so $\ker \gamma_x = n\mathbb{Z}$ for a unique $n \geq 0$ (Theorem 11.4.2). Set $|x| = \infty$ if $n = 0$ and $|x| = n$ otherwise.

**Exercise 11.7.5**

Definitions 11.7.2 and 11.7.4 agree.

**Solution.** If $\ker \gamma_x = 0$, $\gamma_x$ is injective and $\langle x \rangle = \operatorname{im} \gamma_x$ is infinite. If $\ker \gamma_x = n\mathbb{Z}$, $n > 0$, then $n$ is the least positive integer with $x^n = 1$, and $|\langle x \rangle| = n$ by Observation 11.7.3. □

**Exercise 11.7.6**

(a) $|x| = 1$ iff $x = 1$.
(b) $|x| = 2$ iff $1 \neq x = x^{-1}$.
(c) Every nonzero element of $(\mathbb{R}, +)$ has order $\infty$.
(d) A $k$-cycle in $S_n$ has order $k$.

**Solution.** (a),

(b) by Observation 11.7.3.

(c) $na \neq 0$ for $n \neq 0$, $a \neq 0$.

(d) $(i_1 \ldots i_k)^m$ sends $i_1 \mapsto i_{1+m}$ (indices mod $k$), which is $i_1$ iff $k \mid m$; the $k$-th power is $id$. □

**Exercise 11.7.7**

If $|x| = n$ and $a, b \in \mathbb{Z}$, then $x^a = x^b$ iff $n \mid (a - b)$.

**Solution.** $x^a = x^b \iff x^{a-b} = 1 \iff a - b \in \ker \gamma_x = n\mathbb{Z}$. □

**Exercise 11.7.8**

Every subgroup of a cyclic group is cyclic.

**Solution.** Let $G = \langle x \rangle$, $H \leq G$. Then $\gamma_x : \mathbb{Z} \to G$ is onto, $\gamma_x^{-1}(H) \leq \mathbb{Z}$ equals $m\mathbb{Z}$ (Theorem 11.4.2), and $H = \gamma_x(m\mathbb{Z}) = \langle x^m \rangle$. □

**Exercise 11.7.9**

Up to isomorphism the cyclic groups are $\mathbb{Z}$ and the $\mu_n$.

**Solution.** Let $G = \langle x \rangle$. If $\ker \gamma_x = 0$, $\gamma_x : \mathbb{Z} \to G$ is an isomorphism. If $\ker \gamma_x = n\mathbb{Z}$, $n > 0$, $\varphi(x^a) = \zeta^a$ is well defined and injective since $x^a = x^b \iff n \mid a-b \iff \zeta^a = \zeta^b$ (Exercise 11.7.7); it is onto and a homomorphism. □

## 11.8 Examples of groups and homomorphisms

**Motivation.** Symmetric groups, Cayley's theorem, conjugacy, automorphisms and products.

**Exercise 11.8.1**

(a) Composites of homomorphisms are homomorphisms.
(b) If $L \in \mathfrak{L}(V, W)$, then $L : (V, +) \to (W, +)$ is a homomorphism.

**Solution.** (a) $\varphi\psi(xy) = \varphi(\psi x \psi y) = \varphi\psi x \, \varphi\psi y$.

(b) $L(v + w) = Lv + Lw$. □

**Exercise 11.8.2**

(a) The constant map $\mathbf{1}(g) = 1$ is a homomorphism, the **trivial map** (written 0 for abelian groups). (b) For $H \leq G$, the inclusion $\iota_H : H \to G$ is a monomorphism, the **natural embedding**. (c) For $H \leq G$ and a homomorphism $\varphi : G \to G'$, the restriction $\varphi|_H$ is a homomorphism.

**Solution.** All follow directly from the definitions ($1 \cdot 1 = 1$; the operation of $H$ is that of $G$). □

**Example 11.8.3**

$S_n$ is generated by transpositions, and $A_n = \ker(\mathrm{sgn}) \leq S_n$ (Observation 11.6.6).

**Example 11.8.4**

(a) In $S_3$, put $\sigma = (1\,2)$, $\tau = (1\,2\,3)$. Then $\sigma^2 = id = \tau^3$, $\tau^2 = (1\,3\,2)$, $\sigma\tau = (2\,3)$, $\sigma\tau^2 = (1\,3)$, so

$$S_3 = \{id, \tau, \tau^2, \sigma, \sigma\tau, \sigma\tau^2\} = \{\sigma^i\tau^j : 0 \leq i \leq 1,\ 0 \leq j \leq 2\} = \langle \sigma, \tau \rangle.$$

(b) $\sigma^{-1}\tau\sigma = \tau^2 = \tau^{-1}$, i.e. $\tau\sigma = \sigma\tau^2$, moves $\sigma$ to the left, so the relations

$$\sigma^2 = 1, \qquad \tau^3 = 1, \qquad \tau\sigma = \sigma\tau^2$$

determine the whole multiplication table of $S_3$; e.g. $(\sigma\tau)(\sigma\tau^2) = \sigma(\tau\sigma)\tau^2 = \sigma\sigma\tau^2\tau^2 = \tau$.
(c) This is the plane rule of Observation 9.5.9; see Example 11.9.23.

**Definition 11.8.5 (Symmetric group on a set)**

For a nonempty set $X$, $S_X = \{\sigma : X \to X \text{ bijection}\}$ is a group under composition, the **symmetric group** on $X$.

**Exercise 11.8.6**

(a) How should "$X$ and $Y$ are the same as sets" be defined?
(b) If $X, Y$ are isomorphic as sets, then $S_X \approx S_Y$.

**Solution.** (a) There is a bijection $f : X \to Y$ (a renaming).

(b) $\sigma \mapsto f \circ \sigma \circ f^{-1}$ is a bijection $S_X \to S_Y$ with inverse $\tau \mapsto f^{-1}\tau f$, and $f\sigma\sigma' f^{-1} = (f\sigma f^{-1})(f\sigma' f^{-1})$. □

**Exercise 11.8.7**

(a) The **automorphism group** $\mathrm{Aut}(G) = \{\varphi : G \to G \text{ automorphism}\}$ is a subgroup of $S_G$.
(b) For $g \in G$, $\mathrm{Int}(g)(x) = gxg^{-1}$ defines $\mathrm{Int}(g) \in \mathrm{Aut}(G)$, an **inner automorphism**; $\mathrm{Int}(G) = \{\mathrm{Int}(g) : g \in G\} \leq \mathrm{Aut}(G)$.
(c) (Remark) Automorphisms that are not inner are **outer**.

**Solution.** (a) Composites and inverses of automorphisms are automorphisms (Exercise 11.5.6), and $id \in \mathrm{Aut}(G)$. (b) $gxyg^{-1} = gxg^{-1}gyg^{-1}$; inverse $\mathrm{Int}(g^{-1})$; $\mathrm{Int}(g) \circ \mathrm{Int}(h) = \mathrm{Int}(gh)$. □

**Theorem 11.8.8 (Cayley)**

$\lambda : G \to S_G$, $\lambda(x) = \lambda_x$ with $\lambda_x(g) = xg$, is a monomorphism. Hence every group is isomorphic to a subgroup of a symmetric group.

**Proof.** $\lambda_x \in S_G$ (Observation 11.2.15); $\lambda_{xy}(g) = xyg = \lambda_x\lambda_y(g)$; and $\lambda_x = id$ gives $x = \lambda_x(1) = 1$. Apply Observations 11.6.7, 11.6.6(b). ■

(Right multiplication gives $\rho_{xy} = \rho_y \circ \rho_x$, an anti-homomorphism.)

**Definition 11.8.9 (Conjugacy)**

$x \sim y$ iff $y = g^{-1}xg$ for some $g \in G$; this is an equivalence relation, the **conjugacy relation**, and its classes are the **conjugacy classes** ($x$ is **conjugate** to $y$).

**Exercise 11.8.10**

(a) The conjugacy class of 1 is $\{1\}$.
(b) If $G$ is commutative, conjugacy is equality.
(c) How many conjugacy classes does $S_3$ have?

**Solution.** (a) $g^{-1}1g = 1$.

(b) $g^{-1}xg = x$.

(c) Three: $\{id\}$, the transpositions, the 3-cycles (cycle type is preserved; $\sigma^{-1}\tau\sigma = \tau^2$, $\tau^{-1}\sigma\tau = \sigma\tau^2$, ...). □

**Exercise 11.8.11**

(a) For $H \leq G$, $g \in G$: $g^{-1}Hg = \{g^{-1}hg : h \in H\} \leq G$.
(b) On the set $\mathcal{S}$ of subgroups of $G$, $H \sim K$ iff $K = g^{-1}Hg$ for some $g$ is an equivalence relation ($K$ is a **conjugate subgroup** of $H$).

**Solution.** (a) $g^{-1}Hg = \mathrm{Int}(g^{-1})(H)$, the image of a subgroup under a homomorphism.

(b) Reflexive with $g = 1$; symmetric since $H = gKg^{-1}$; transitive since $h^{-1}(g^{-1}Hg)h = (gh)^{-1}H(gh)$. □

**Exercise 11.8.12**

(a) Give the Cartesian product $G = G_1 \times \cdots \times G_n$ of groups a natural group structure. (b) The $i$-th **projection** $\pi_i : G \to G_i$ is an epimorphism and the $j$-th **embedding** $\iota_j : G_j \to G$ is a monomorphism.

**Solution.** (a) Componentwise multiplication $(g_i)(h_i) = (g_ih_i)$; identity $(1, \ldots, 1)$, inverse $(g_i^{-1})$.
(b) $\pi_i(g) = g_i$; $\iota_j(g) = (1, \ldots, g, \ldots, 1)$ ($g$ in position $j$). □

**Exercise 11.8.13**

(a) $G \times H \approx H \times G$.
(b) $(G \times H) \times K \approx G \times (H \times K) \approx G \times H \times K$.

**Solution.** $(g, h) \mapsto (h, g)$ and $((g, h), k) \mapsto (g, h, k)$ and $(g, (h, k)) \mapsto (g, h, k)$ are bijective homomorphisms. □

**Exercise 11.8.14**

$G_1 \times \cdots \times G_n$ is commutative iff all $G_i$ are commutative.

**Solution.** Componentwise; conversely $G_i \approx \iota_i(G_i)$ is a subgroup. □

**Exercise 11.8.15**

(a) $\mu_3 \times \mu_3 \not\approx \mu_9$.
(b) $\mu_2 \times \mu_3 \approx \mu_6$.

**Solution.** (a) Elements of $\mu_3 \times \mu_3$ have order dividing 3; $\mu_9$ has one of order 9.
(b) $(-1, \omega)^k = 1$ ($\omega = e^{2\pi i/3}$) iff $2 \mid k$ and $3 \mid k$, so it has order 6 and $\mu_2 \times \mu_3 \approx \mu_6$ (Exercise 11.7.9). □

**Exercise 11.8.16**

In Exercise 11.8.12(b): (a) find $\ker \pi_i$ and $\operatorname{im} \iota_j$;
(b) $\pi_i \circ \iota_i = id$;
(c) $\pi_i \circ \iota_j = \mathbf{1}$ for $i \neq j$.

**Solution.** (a) $\ker \pi_i = \{g : g_i = 1\}$ and $\operatorname{im} \iota_j = \{g : g_k = 1 \text{ for } k \neq j\}$. (b), (c) are direct. □

**Remark 11.8.17**

The pointwise product $(\varphi\psi)(g) = \varphi(g)\psi(g)$ of homomorphisms need not be one: $\varphi(x)\varphi(y)\psi(x)\psi(y) \neq \varphi(x)\psi(x)\varphi(y)\psi(y)$ unless $H$ is commutative.

## 11.9 Linear groups

**Motivation.** A basis identifies $\mathrm{GL}(V)$ with $\mathrm{GL}_n(F)$, and every finite group is a matrix group.

**Example 11.9.1**

(a) $\mathrm{GL}_n(F) = \{A \in \mathfrak{M}_{n,n}(F) : A \text{ invertible}\}$, the **general linear group**, is a group under matrix multiplication; $F^\times = \mathrm{GL}_1(F) = F - \{0\}$ is the **unit** (multiplicative) **group** of $F$.
(b) Subgroups of $\mathrm{GL}_n(F)$ are **linear** (**matrix**) **groups**.
(c) The **special linear group** $\mathrm{SL}_n(F) = \{A \in \mathrm{GL}_n(F) : \det A = 1\} = \ker(\det : \mathrm{GL}_n(F) \to F^\times)$ is a subgroup.

**Example 11.9.2**

For a finite-dimensional $F$-vector space $V$: (a) $\mathrm{GL}(V) = \{L \in \mathfrak{L}(V,V) : L \text{ invertible}\}$ is a group under composition; (b) its subgroups are **linear groups**;
(c) $\mathrm{SL}(V) = \{L \in \mathrm{GL}(V) : \det L = 1\} \leq \mathrm{GL}(V)$.

**Example 11.9.3**

(a) For $\dim V = n$ and a basis $\mathfrak{B}$, the vector space isomorphism $\Psi_{\mathfrak{B}}^{\mathfrak{B}} : \mathfrak{L}(V,V) \to \mathfrak{M}_{n,n}(F)$, $L \mapsto [L]_{\mathfrak{B}}^{\mathfrak{B}}$, restricts to a group isomorphism $\mathrm{GL}(V) \to \mathrm{GL}_n(F)$ ($[LM] = [L][M]$).

$$\begin{array}{ccc} \mathrm{GL}(V) & \hookrightarrow & \mathfrak{L}(V,V) \\ {\scriptstyle \Psi_{\mathfrak{B}}^{\mathfrak{B}}}\downarrow & & \downarrow{\scriptstyle \Psi_{\mathfrak{B}}^{\mathfrak{B}}} \\ \mathrm{GL}_n(F) & \hookrightarrow & \mathfrak{M}_{n,n}(F) \end{array}$$

(c) Since $\det L = \det[L]_{\mathfrak{B}}^{\mathfrak{B}}$, it also restricts to $\mathrm{SL}(V) \approx \mathrm{SL}_n(F)$.

**Example 11.9.4**

If $\dim V = \dim W = n$, then $\mathrm{GL}(V) \approx \mathrm{GL}(W)$. (a) Via bases: $(\Psi_{\mathfrak{C}}^{\mathfrak{C}})^{-1} \circ \Psi_{\mathfrak{B}}^{\mathfrak{B}}$. (b) Via a vector space isomorphism $\varphi : V \to W$ (e.g. with $\varphi(\mathfrak{B}) = \mathfrak{C}$): $\Gamma(L) = \varphi \circ L \circ \varphi^{-1}$, with $\Gamma(LL') = \Gamma(L)\Gamma(L')$ and $\Gamma^{-1}(M) = \varphi^{-1} M \varphi$. (c) If $\varphi(\mathfrak{B}) = \mathfrak{C}$ they coincide: $[\varphi L \varphi^{-1}]_{\mathfrak{C}}^{\mathfrak{C}} = [\varphi]_{\mathfrak{B}}^{\mathfrak{C}}[L]_{\mathfrak{B}}^{\mathfrak{B}}[\varphi^{-1}]_{\mathfrak{C}}^{\mathfrak{B}} = [L]_{\mathfrak{B}}^{\mathfrak{B}}$, as $[\varphi]_{\mathfrak{B}}^{\mathfrak{C}} = I$.

**Example 11.9.5**

$\mu_2 < \mathbb{R}^\times < \mathbb{C}^\times$ and $\mu_n < \mathbb{C}^\times$.

**Example 11.9.6**

Theorem 1.3.6(3) (every invertible matrix is a product of elementary matrices) says: $\mathrm{GL}_n(F)$ is generated by elementary matrices.

**Example 11.9.7**

$\mathbf{T}_n(F) = \{A \in \mathrm{GL}_n(F) : A \text{ upper triangular}\}$ and $\mathbf{V}_n(F) = \{A \in \mathbf{T}_n(F) : A \text{ unipotent}\}$ (diagonal entries 1) are matrix groups.

**Proof [Supplement].** Products and inverses (back substitution) of invertible upper triangular matrices are upper triangular, with diagonal entries multiplied resp. inverted. ■

**Exercise 11.9.8**

$(F,+)$ is a matrix group: $(F,+) \approx \mathbf{V}_2(F)$.

**Solution.** $a \mapsto \left(\begin{smallmatrix}1 & a\\ 0 & 1\end{smallmatrix}\right)$ is a bijection, and $\left(\begin{smallmatrix}1 & a\\ 0 & 1\end{smallmatrix}\right)\left(\begin{smallmatrix}1 & b\\ 0 & 1\end{smallmatrix}\right) = \left(\begin{smallmatrix}1 & a+b\\ 0 & 1\end{smallmatrix}\right)$. □

**Exercise 11.9.9**

The group $(F^3,\bullet)$ of Exercise 11.1.14 is isomorphic to $\mathbf{V}_3(F)$.

**Solution.** $(a_1,a_2,a_3)^t \mapsto \begin{pmatrix}1 & a_1 & a_2\\ 0 & 1 & a_3\\ 0 & 0 & 1\end{pmatrix}$; the product has entries $a_1+b_1$, $a_2+b_2+a_1b_3$, $a_3+b_3$, the formula for $\bullet$. □

**Example 11.9.10**

Conjugacy in $\mathrm{GL}_n(F)$ is the similarity relation of §5.5.

**Exercise 11.9.11**

If $L \in \mathrm{GL}(V)$ is unipotent and $L \neq I$, then $|L| = \infty$ in $\mathrm{GL}(V)$ (over $F = \mathbb{R}$ or $\mathbb{C}$).

**Solution.** $L = I + N$, $0 \neq N$ nilpotent, $k \geq 1$ largest with $N^k \neq 0$. Then $L^m - I = \sum_{1\leq j\leq k} \binom{m}{j} N^j$ with coefficient $m \neq 0$ at $N$, and $I, N, \ldots, N^k$ are independent (for $\sum c_j N^j = 0$ with first nonzero $c_r$, multiply by $N^{k-r}$: $c_r N^k = 0$). So $L^m \neq I$ for $m \geq 1$. □

**Exercise 11.9.12**

For fixed $C \in \mathrm{GL}_n(\mathbb{R})$, $\varphi(A) = CAC^{-1}$ and $\psi(A) = (A^t)^{-1}$ belong to $\mathrm{Aut}(\mathrm{SL}_n(\mathbb{R}))$. If $C \notin \mathrm{SL}_n(\mathbb{R})$, are $\varphi, \psi$ outer automorphisms?

**Solution.** Both preserve $\mathrm{SL}_n(\mathbb{R})$ (det is unchanged) and products ($((AB)^t)^{-1} = (A^t)^{-1}(B^t)^{-1}$), and are bijective ($\psi^2 = id$, $\varphi^{-1} = \mathrm{Int}(C^{-1})$).

[Supplement] $\varphi$. $C' \in \mathrm{SL}_n(\mathbb{R})$ induces $\varphi$ iff $C^{-1}C'$ commutes with $\mathrm{SL}_n(\mathbb{R})$ (all type-3 elementary matrices), i.e. is $\lambda I$. So $\varphi$ is inner iff $\lambda^n \det C = 1$ for some real $\lambda$: always for odd $n$; never for even $n$ with $\det C < 0$ (outer).

$\psi$. For $n = 2$, $(A^t)^{-1} = JAJ^{-1}$ with $J = \left(\begin{smallmatrix}0 & 1\\ -1 & 0\end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{R})$: inner. For $n \geq 3$, outer, since inner automorphisms preserve trace: $A = \mathrm{diag}(2, 3, \frac{1}{6}, 1, \ldots, 1)$ has trace $5\frac{1}{6} + (n-3)$, $\psi(A) = \mathrm{diag}(\frac{1}{2}, \frac{1}{3}, 6, 1, \ldots, 1)$ has trace $6\frac{5}{6} + (n-3)$. □

**Exercise 11.9.13**

For the invertible diagonal matrices $\mathbf{D}_n(F)$: (a) $\mathbf{D}_n(F) \approx F^\times \times \cdots \times F^\times$ ($n$ factors);
(b) $\mathbf{D}_n(F) \cap \mathrm{SL}_n(F) \approx (F^\times)^{n-1}$ for $n \geq 2$.

**Solution.** (a) $\mathrm{diag}(a_1, \ldots, a_n) \mapsto (a_1, \ldots, a_n)$.

(b) The last entry is determined by $a_n = (a_1 \cdots a_{n-1})^{-1}$; project to the first $n-1$ entries. □

**Exercise 11.9.14**

(a) If $m \leq n$, $\mathrm{GL}_m(F)$ is isomorphic to a subgroup of $\mathrm{GL}_n(F)$.
(b) If $n_1 + \cdots + n_r = n$, $\mathrm{GL}_{n_1}(F) \times \cdots \times \mathrm{GL}_{n_r}(F)$ is isomorphic to a subgroup of $\mathrm{GL}_n(F)$.

**Solution.** (b) $(A_1, \ldots, A_r) \mapsto A_1 \oplus \cdots \oplus A_r$ (block diagonal) is a monomorphism. (a) is the case $A \mapsto A \oplus I_{n-m}$. □

**Exercise 11.9.15**

(a) $\mathbf{SO}(n) < \mathbf{O}(n) < \mathrm{GL}_n(\mathbb{R})$;
(b) $\mathbf{SU}(n) < \mathbf{U}(n) < \mathrm{GL}_n(\mathbb{C})$;
(c) $\mathbf{SO}(n) = \mathrm{SL}_n(\mathbb{R}) \cap \mathbf{O}(n)$;
(d) $\mathbf{SU}(n) = \mathrm{SL}_n(\mathbb{C}) \cap \mathbf{U}(n)$.

**Solution.** By Exercises 9.3.3, 9.3.10, 10.6.10 and the definitions. □

**Example 11.9.16**

$\Psi_{\mathcal{E}}^{\mathcal{E}}$ restricts to $\mathbf{O}(\mathbb{R}^n) \approx \mathbf{O}(n)$ (dot product), and so does $\Psi_{\mathfrak{B}}^{\mathfrak{B}}$ for any orthonormal $\mathfrak{B}$ (Remark 10.6.4, Exercise 9.6.11).

**Exercise 11.9.17**

(a) $\mathbf{U}(\mathbb{C}^n) \approx \mathbf{U}(n)$.
(b) $\mathbf{SO}(\mathbb{R}^n) \approx \mathbf{SO}(n)$.
(c) Define $\mathbf{SU}(\mathbb{C}^n)$ and show $\mathbf{SU}(\mathbb{C}^n) \approx \mathbf{SU}(n)$.

**Solution.** As in Example 11.9.16, restricting $\Psi_{\mathcal{E}}^{\mathcal{E}}$; for (b), (c) also $\det L = \det[L]_{\mathcal{E}}^{\mathcal{E}}$, with $\mathbf{SU}(\mathbb{C}^n) = \{L \in \mathbf{U}(\mathbb{C}^n) : \det L = 1\}$. □

**Example 11.9.18**

If $\varphi : V \to W$ is an inner product space isomorphism, $\Gamma(L) = \varphi L \varphi^{-1}$ (Example 11.9.4) restricts to $\mathbf{O}(V) \approx \mathbf{O}(W)$ (resp. $\mathbf{U}$), since $L$ preserves inner products iff $\varphi L \varphi^{-1}$ does.

**Example 11.9.19**

The **circle group** $S^1 = \{X \in \mathbb{R}^2 : \|X\| = 1\} = \{z \in \mathbb{C} : |z| = 1\} \leq \mathbb{C}^\times$, and $\varphi : \mathbf{SO}(2) \to S^1$, $\varphi(R_\theta) = e^{i\theta}$, is a group isomorphism.

**Exercise 11.9.20**

$\mathbf{SO}(2) \approx S^1 \approx \mathbf{U}(1)$.

**Solution.** $\varphi(R_\theta R_\eta) = \varphi(R_{\theta+\eta}) = e^{i(\theta+\eta)} = e^{i\theta}e^{i\eta}$, and $\varphi$ is bijective (Theorem 9.5.2). $\mathbf{U}(1) = \{z \in \mathbb{C} : \bar{z}z = 1\} = S^1$. □

**Exercise 11.9.21**

$S_n \approx \{I_\sigma \in \mathfrak{M}_{n,n}(F) : \sigma \in S_n\} \approx \{P_\sigma \in \mathfrak{L}(V,V) : \sigma \in S_n\}$ ($\dim V = n$), via permutation matrices and permutation operators.

**Solution.** The permutation matrix $I_\sigma$ (with $I_\sigma e_j = e_{\sigma(j)}$) satisfies $I_\sigma I_\tau = I_{\sigma\tau}$, and $\sigma \mapsto I_\sigma$ is injective; similarly for $P_\sigma$ on a basis. □

**Corollary 11.9.22**

Every finite group is isomorphic to a linear (matrix) group.

**Proof.** By Cayley's theorem $G$ embeds in $S_G \approx S_n$ ($n = |G|$), and $S_n$ embeds in $\mathrm{GL}_n(F)$ by Exercise 11.9.21. ■

**Example 11.9.23**

In $\mathbf{O}(2)$ let $R = R_{2\pi/3}$, $S = \left(\begin{smallmatrix} -1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, and $D_3 = \langle R, S\rangle \leq \mathbf{O}(2)$.
(a) The plane rule gives $RS = SR^2$. $D_3$ consists of words in $R, R^2, S$; moving each $S$ left gives $D_3 = \{S^i R^j : 0 \leq i \leq 1,\ 0 \leq j \leq 2\}$, six distinct matrices (determinants, first columns). $S^2 = I$, $R^3 = I$, $RS = SR^2$ determine the table; e.g. $(SR)(SR^2) = S(RS)R^2 = S^2R^4 = R$.
(b) These are the relations $\sigma^2 = 1$, $\tau^3 = 1$, $\tau\sigma = \sigma\tau^2$ of $S_3$ (Example 11.8.4).
(c) So $\sigma^i\tau^j \mapsto S^iR^j$ is a table-preserving bijection: $S_3 \approx D_3$.

**Exercise 11.9.24**

For $n \geq 3$, the **dihedral group** $D_n = \langle R_{2\pi/n}, S\rangle \leq \mathbf{O}(2)$ has $|D_n| = 2n$.

**Solution.** $R^n = I$, $S^2 = I$, $RS = SR^{-1}$ give $D_n = \{S^iR^j : 0 \leq i \leq 1, 0 \leq j < n\}$ as in Example 11.9.23; distinct since $\det(S^iR^j) = (-1)^i$ and the $R^j$ are distinct. □

**Exercise 11.9.25**

(a) For $\sigma = (2\,5)(3\,4)$, $\tau = (1\,2\,3\,4\,5) \in S_5$, $G_5 = \langle\sigma,\tau\rangle \approx D_5$. (b) For $\sigma = (2\,6)(3\,5)$, $\tau = (1\,2\,3\,4\,5\,6) \in S_6$, $G_6 = \langle\sigma,\tau\rangle \approx D_6$. (c) Interpret (a),
(b) as permutations of the vertices of a regular pentagon/hexagon, and generalize to the regular $n$-gon.

**Solution.** $\sigma^2 = 1$, $\tau^n = 1$, $\sigma\tau\sigma = \tau^{-1}$ (e.g. $\sigma\tau\sigma(1) = \sigma(2) = n = \tau^{-1}(1)$), so $\sigma^i\tau^j \mapsto S^iR^j$ is an isomorphism onto $D_n$ (Example 11.9.23). With vertices of the regular $n$-gon numbered counterclockwise, $\tau$ is rotation by $2\pi/n$ and $\sigma$ the reflection through vertex 1: $D_n$ is the symmetry group of the $n$-gon. □